Chapter 10 Normal subgroups and Factor groups.

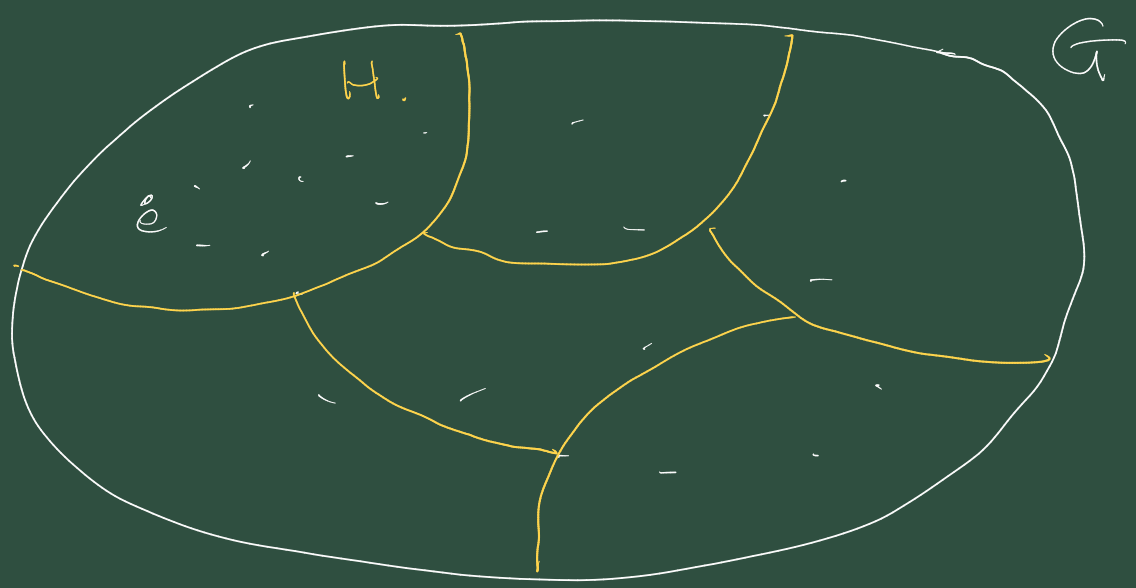
Approach to the general question of asking how complicated a group is?.

Def of a "<u>normal</u> subgroup N of a group G" is that

$$\forall g \in G \quad gN = Ng$$

Motivating this definition.

Given a subgroup H of a group G



Can partition G into the set of left cosets of H in G.

- This can be considered a simplified view of G, with reference to H.
- Can this partition be thought of as a group itself?

What might the operation be among the cosets?

What ways of "multiplying" cosets might there be?

- $\boxed{(aH)(bH) = (\underline{ab})H}$

  One possible definition for multiplying cosets.

- More generally, given two subsets $X, Y \subset G$ of a group G, we can always ~~the~~ carry out the set-wise product.

  $$XY = \{xy : x \in X, y \in Y\} \subset G.$$

Things might work out well if these two definitions coincide. That is, let's investigate how.

$(aH)(bH)$, the set-wise product.

$= \{ a h_1 b h_2 : h_1, h_2 \in H \}$

$\overset{?}{=} abH.$

$\iff \forall h_1 \in H \; \forall h_2 \in H \; \exists h_3 \in H$ such that

$$a h_1 b h_2 = ab h_3$$

$\iff \forall h_1, h_2 \in H \; \exists h_3 \in H.$

$$h_1 b h_2 = b h_3$$

$\iff$

$$h_1 b = b h_3 h_2^{-1}$$

$\iff$ $\boxed{\begin{array}{c} \forall h_1 \in H. \; \exists h_4 \in H \\ h_1 b = b h_4 \end{array}}$ (just writing $h_4 = h_3 h_2^{-1}$)

This says the right coset Hb is equal to the left coset bH.

This gives us the definition of normality of H in G.

<u>Ex 10.1</u> If G is abelian every subgroup H is normal in G, as $\forall\, g, h \in H$.

$hg = gh$

$\Rightarrow Hg = gH$.

<u>Ex 10.2</u> $G = S_3$

$H = \{(1), (1\,2)\}$

$(1\,2\,3)H = \{(1\,2\,3), (1\,3)\}$

$H(1\,2\,3) = \{(1\,2\,3), (2\,3)\}$.

different subsets.

So this H is <u>not</u> normal in S_3.

But the subgroup $N = \{(1), (1\,2\,3), (1\,3\,2)\}$ is normal. There are only two cosets

N, $(1\,2)N = N(1\,2)$

Theorem 10.3

Equivalent versions of the definition.

1. N is normal in G (*)
2. $\forall g \in G \quad gNg^{-1} \subset N$
3. $\forall g \in G \quad \underbrace{gNg^{-1}}_{\text{"conjugation of } N \text{ by } g\text{"}} = N.$ (*)

$$gNg^{-1} = \{ gng^{-1} : n \in N \}$$

So when H is not normal in G then for some $g \in G$ gHg^{-1} will not be equal to H, it will be another subgroup of size $|H|$.

Theorem 10.4

Given a normal subgroup N of a group G. The collection of cosets of N in G forms a group under the operation.

$$(aN)(bN) := \underline{\underline{(ab)N}} \quad \text{(quotient)}$$

This group is called the factor group of G by N, denoted G/N.

Proof. First, caution, we shall show the operation is well defined, that it does not make a difference if we use different representatives of aN and bN.

Let $a, b, c, d \in G$ satisfying.

$aN = cN$, $bN = dN$

$\boxed{\text{Claim. } (ab)N = (cd)N}$

(Examining Lemma 6.3, and looking for the most suitable condition, choose prop 4)

Note $g_1 H = g_2 H \iff g_2 \in g_1 H$.

$aN = cN \iff c \in aN \iff c = an$, for some $n \in N$.

$bN = dN \iff d \in bN \iff d = bm$, for some $m \in N$

We want to show that $\boxed{cd \in abN}$.

$\boxed{cd} = anbm$

$= abn'm$, for some $n' \in N$ since N is normal, so $Nb = bN$.

$\boxed{\in (ab)N}$, since $n'm \in N$, as N is a subgroup.

So by Lemma 6.3 $(ab)N = (cd)N$

So the operation is well-defined.

(Pause continue at 12:05).

Now we prove the necessary group conditions.

- The operation is self-evidently closed on G/N, the set of cosets.
- Claim: the operation is associative.

Consider.

$$\boxed{aN(bN\ cN)} = \underbrace{aN}\ \underbrace{(bc)N}\ , \text{ by def.}$$

$$= (\underbrace{a(bc)})N\ , \text{ by def.}$$

$$= ((ab)c)N\ , \text{ associativity of op in } G.$$

$$= (ab)N\ cN\ , \text{ by def.}$$

$$= (aN\ bN)\ cN\ , \text{ by def}$$

$$= \cancel{(aN\ bN)\ cN}$$

This shows the op. is associative on G/N.

The identity is for this new operation is the coset eN, or just N.

since.

$$(aN)(eN) = (ae)N = aN$$

and similarly.

$$(eN)(aN) = (ea)N = aN$$

Finally, do all inverses exist?

Let $aN \in G/N$

the inverse for this is $(aN)^{-1} = \underline{a^{-1}N}$

since

$$(aN)(a^{-1}N) = (aa^{-1})N$$
$$= eN = N,$$ the identity

element for G/N.

So <u>all</u> the group conditions hold, so G/N is indeed a group ∎

Ex 10.5. For $N = \{(1), (123), (132)\}$ the normal subgroup of S_3, S_3/N consists of N, $(12)N = \{(12), (13), (23)\}$.

The Cayley table for this factor group.

S_3/N	N	$(12)N$
N	N	$(12)N$
$(12)N$	$(12)N$	N

since $(12)N(12)N = (12)(12)N$
$= N$

Compare this Cayley table with the one for $\mathbb{Z}_2$.

	0	1
0	0	1
1	1	0

So $S_3/N \cong \mathbb{Z}_2$.

the isomorphism being $\phi : S_3/N \to \mathbb{Z}_2$.

$$N \longmapsto 0$$

$$(12)N \longmapsto 1.$$

<u>Example 10.6</u>.

Let $G = \mathbb{Z}$, the integers under addition.

Consider the subgroup $3\mathbb{Z}$.

$$3\mathbb{Z} = \{\ldots, -9, -6, -3, 0, 3, 6, 9, \ldots\}.$$

~~the other~~ $\mathbb{Z}$ is abelian, so $3\mathbb{Z}$ is automatically normal. The other non-identity cosets are.

$$1 + 3\mathbb{Z} = \{\ldots, -8, -5, -2, 1, 4, 7, 10, \ldots\}$$

$$2 + 3\mathbb{Z} = \{\ldots, -7, -4, -1, 2, 5, 8, 11, \ldots\}.$$

Examining the Cayley table for the factor group $\mathbb{Z}/3\mathbb{Z}$ (see example 10.6. in Judson).

It is isomorphic

	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

which is exactly the Cayley table for $\mathbb{Z}_3$, the integers modulo 3

under addition.

So $\mathbb{Z}/3\mathbb{Z}$ <u>is</u> the group $\mathbb{Z}_3$

and in general

$\mathbb{Z}/n\mathbb{Z}$ is the group $\mathbb{Z}_n$

So can think of Factor group construction as a generalization of the concepts of modular arithmetic to other settings.

<u>Example 10.7</u>. $G = D_n$, the group of symmetries of a regular n-gon, n-sided regular polygon.

$$D_n = \langle r, s : r^n = e, s^2 = e, sr = r^{-1}s \rangle$$
$$= \{ e, r, r^2, r^3, \ldots, r^{n-1}, s, rs, r^2s, \ldots, r^{n-1}s \}.$$

$|D_n| = 2n$

Consider $R = \langle r \rangle$, the subgroup of rotations from D_n. This is normal, there are two only two cosets R, sR.

So D_n/R is again a two-element group, and so isomorphic to $\mathbb{Z}_2$.

	R	sR
R	R	sR
sR	sR	R.